

PROJECT REPORT

REAL-TIME STOCK & STOCK FORECAST DASHBOARD



Carried Out at
CENTRE FOR DEVELOPMENT OF ADVANCED COMPUTING
ELECTRONIC CITY, BANGALORE

UNDER THE SUPERVISION OF
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Candidate's Declaration

We hereby certify that the work being presented in the report entitled "**Real-Time Stock & Stock Forecast Dashboard**," in partial fulfillment of the requirements for the award of PG Diploma in Big Data Analytics and submitted to the department of Big Data Analytics, C-DAC Bangalore, is an authentic record of our work carried out under the supervision of Mr. Prasoon P, C-DAC Bangalore.

The matter presented in the report has not been submitted by me for the award of any degree of this or any other Institute/University.

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ABSTRACT

Financial markets play a crucial role in the global economy. Tracking and forecasting stock prices help investors make informed decisions. This project focuses on developing a **Real-Time Stock & Stock Forecast Dashboard** using **Streamlit**, **Plotly**, **Yahoo Finance (yfinance)**, and **Prophet** for stock price prediction. The dashboard allows users to monitor live stock prices, visualize trends, and forecast future stock movements. The project leverages technical indicators such as **SMA (Simple Moving Average)**, **EMA (Exponential Moving Average)**, and **Percentage Change Forecasting** to enhance analytical capabilities.

This report details the system design, implementation, and results, highlighting the effectiveness of data-driven financial analysis. The project also explores the integration of machine learning models for more accurate predictions and discusses potential future enhancements.

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CHAPTER 1: INTRODUCTION

1.1 Overview

The stock market is one of the most dynamic and complex systems in the financial world. Investors, traders, and analysts rely on accurate and timely information to make informed decisions. With the advent of **Big Data** and **Machine Learning**, the ability to predict stock prices and monitor market trends has significantly improved. This project aims to develop a **Real-Time Stock & Stock Forecast Dashboard** that provides users with live stock price updates, interactive visualizations, and predictive analytics using advanced machine learning models.

1.2 Objectives

- Develop a **real-time stock price dashboard** using **Streamlit**.
- Implement **technical indicators** such as **SMA (Simple Moving Average)** and **EMA (Exponential Moving Average)**.
- Enable **stock price forecasting** using **Facebook's Prophet**.
- Provide users with **interactive visualizations** using **Plotly**.
- Enhance the dashboard with **machine learning models** for more accurate predictions.

1.3 Problem Statement

The stock market is highly volatile, and predicting stock prices is a challenging task. Traditional methods of stock analysis often fail to capture the non-linear patterns in stock price movements. This project aims to address these challenges by developing a real-time dashboard that not only monitors stock prices but also provides accurate forecasts using advanced machine learning techniques.

1.4 Scope of the Project

The scope of this project includes:

- Real-time stock price monitoring using **Yahoo Finance API**.
- Implementation of technical indicators such as **SMA**, **EMA**, and **Percentage Change**.
- Stock price forecasting using **Prophet**.
- Development of an interactive dashboard using **Streamlit** and **Plotly**.
- Integration of machine learning models for enhanced forecasting accuracy.

1.5 Challenges

- **Data Latency**: Ensuring real-time data retrieval from Yahoo Finance.
- **Model Accuracy**: Achieving high accuracy in stock price predictions using Prophet.
- **User Experience**: Designing an intuitive and user-friendly dashboard.
- **Scalability**: Ensuring the system can handle multiple users and large datasets.

CHAPTER 2: LITERATURE SURVEY

2.1 Traditional Models for Stock Prediction

Traditional models such as **ARIMA (AutoRegressive Integrated Moving Average)** and **GARCH (Generalized Autoregressive Conditional Heteroskedasticity)** have been widely used for stock price

prediction. However, these models are limited in their ability to capture non-linear patterns in stock price movements.

2.2 Modern Approaches in Stock Forecasting

Modern approaches leverage **machine learning** and **deep learning** techniques to improve prediction accuracy. Models such as **LSTM (Long Short-Term Memory)** and **Prophet** have shown significant improvements in forecasting stock prices. Prophet, developed by Facebook, is particularly effective for time series forecasting due to its ability to handle seasonality and trends.

2.3 Role of Technical Indicators

Technical indicators such as **Moving Averages (SMA, EMA)**, **RSI (Relative Strength Index)**, and **MACD (Moving Average Convergence Divergence)** play a crucial role in stock market analysis. These indicators help traders identify trends, momentum, and potential reversal points in stock prices.

2.4 Dashboards in Financial Analysis

Dashboards have become an essential tool for financial analysis. They provide a centralized platform for monitoring real-time data, visualizing trends, and making data-driven decisions. **Streamlit** has emerged as a popular framework for building interactive financial dashboards due to its simplicity and flexibility.

CHAPTER 3: SOFTWARE REQUIREMENT SPECIFICATION

3.1 Technologies Used

- **Programming Language:** Python
- **Framework:** Streamlit
- **Libraries:** yfinance, Plotly, Prophet, Pandas, NumPy, TA-Lib
- **IDE:** Jupyter Notebook, PyCharm, VS Code

3.2 Hardware Requirements

- **Minimum 8GB RAM**
- **Intel i5 or higher processor**
- **Stable Internet Connection**

3.3 Software Requirements

- **Operating System:** Windows 10, macOS, or Linux
 - **Python Version:** 3.7 or higher
 - **Dependencies:** Streamlit, yfinance, Plotly, Prophet, Pandas, NumPy, TA-Lib
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CHAPTER 4: ARCHITECTURE

4.1 System Overview

The system consists of two main modules:

1. **Real-Time Stock Price Monitoring:** Fetches stock data using **yfinance**, processes it, and visualizes real-time trends.
2. **Stock Forecasting:** Uses **Prophet** to predict stock price movements.

4.2 Data Flow Diagram

1. **User selects a stock** → 2. **System fetches data from Yahoo Finance** → 3. **Preprocesses & visualizes data** → 4. **Forecasts using Prophet** → 5. **Displays results on dashboard**

4.3 System Architecture Diagram

The system architecture is divided into four main components:

1. **Stock Data Retrieval Module:** Fetches stock data from **Yahoo Finance**.
 2. **Data Processing Module:** Applies transformations like SMA, EMA.
 3. **Visualization Module:** Uses **Plotly** for dynamic stock charts.
 4. **Forecasting Module:** Utilizes **Prophet** for future trend predictions.
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CHAPTER 5: SYSTEM DESIGN

5.1 Stock Data Retrieval Module

This module is responsible for fetching real-time stock data from **Yahoo Finance** using the **yfinance** library. The data includes stock prices, volume, and other relevant metrics.

5.2 Data Processing Module

The data processing module applies technical indicators such as **SMA**, **EMA**, and **Percentage Change** to the fetched stock data. These indicators help in identifying trends and patterns in stock prices.

5.3 Visualization Module

The visualization module uses **Plotly** to create interactive charts such as **candlestick charts**, **line charts**, and **bar charts**. These visualizations help users understand stock price movements and trends.

5.4 Forecasting Module

The forecasting module uses **Prophet** to predict future stock prices. Prophet is a time series forecasting model developed by Facebook that can handle seasonality, trends, and holidays.

5.5 User Interface Design

The user interface is designed using **Streamlit**, which provides a simple and intuitive way to interact with the dashboard. The UI includes features such as stock selection, time range selection, and visualization options.

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CHAPTER 6: IMPLEMENTATION

6.1 Real-Time Stock Data Fetching

The system fetches real-time stock data using the **yfinance** library. The data is then processed and stored in a Pandas DataFrame for further analysis.

6.2 Technical Indicators Implementation

Technical indicators such as **SMA**, **EMA**, and **Percentage Change** are implemented using the **TA-Lib** library. These indicators are calculated and displayed on the dashboard.

6.3 Stock Price Forecasting with Prophet

The **Prophet** model is used to forecast future stock prices. The model is trained on historical stock data and generates predictions for the selected time range.

6.4 Dashboard Deployment

The dashboard is deployed using **Streamlit**, which provides a web-based interface for users to interact with the system. The dashboard is hosted on a local server and can be accessed via a web browser.

6.5 Key Features of the Dashboard

- **Live Stock Price Tracking:** Users can monitor real-time stock prices.
 - **Candlestick & Line Charts:** Interactive charts for visualizing stock price movements.
 - **Technical Indicator Visualization:** Display of SMA, EMA, and other indicators.
 - **Future Stock Price Forecasting:** Predictions using Prophet.
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CHAPTER 7: RESULTS AND DISCUSSION

7.1 Real-Time Stock Monitoring

The dashboard successfully fetches and displays real-time stock prices. Users can select different stocks and time ranges to view historical and current data.

7.2 Forecasting Accuracy

The **Prophet** model provides accurate predictions for stock prices. The model's ability to handle seasonality and trends makes it suitable for stock price forecasting.

7.3 User Experience and Feedback

The dashboard is user-friendly and provides a seamless experience for monitoring and analyzing stock prices. Users can easily navigate through different features and visualize data.

CHAPTER 8: CONCLUSION

8.1 Summary of the Project

This project successfully integrates **real-time stock analysis** with **forecasting capabilities**. The dashboard enables traders and analysts to **monitor market trends**, apply **technical indicators**, and **predict stock movements** effectively.

8.2 Future Scope

- Integration with **Deep Learning models** for more accurate forecasting.
- Expanding the dataset to include **global stock markets**.
- Enhancing UI with **interactive widgets** for better user experience.
- Adding more technical indicators and advanced analytics features.

CHAPTER 9: REFERENCES

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